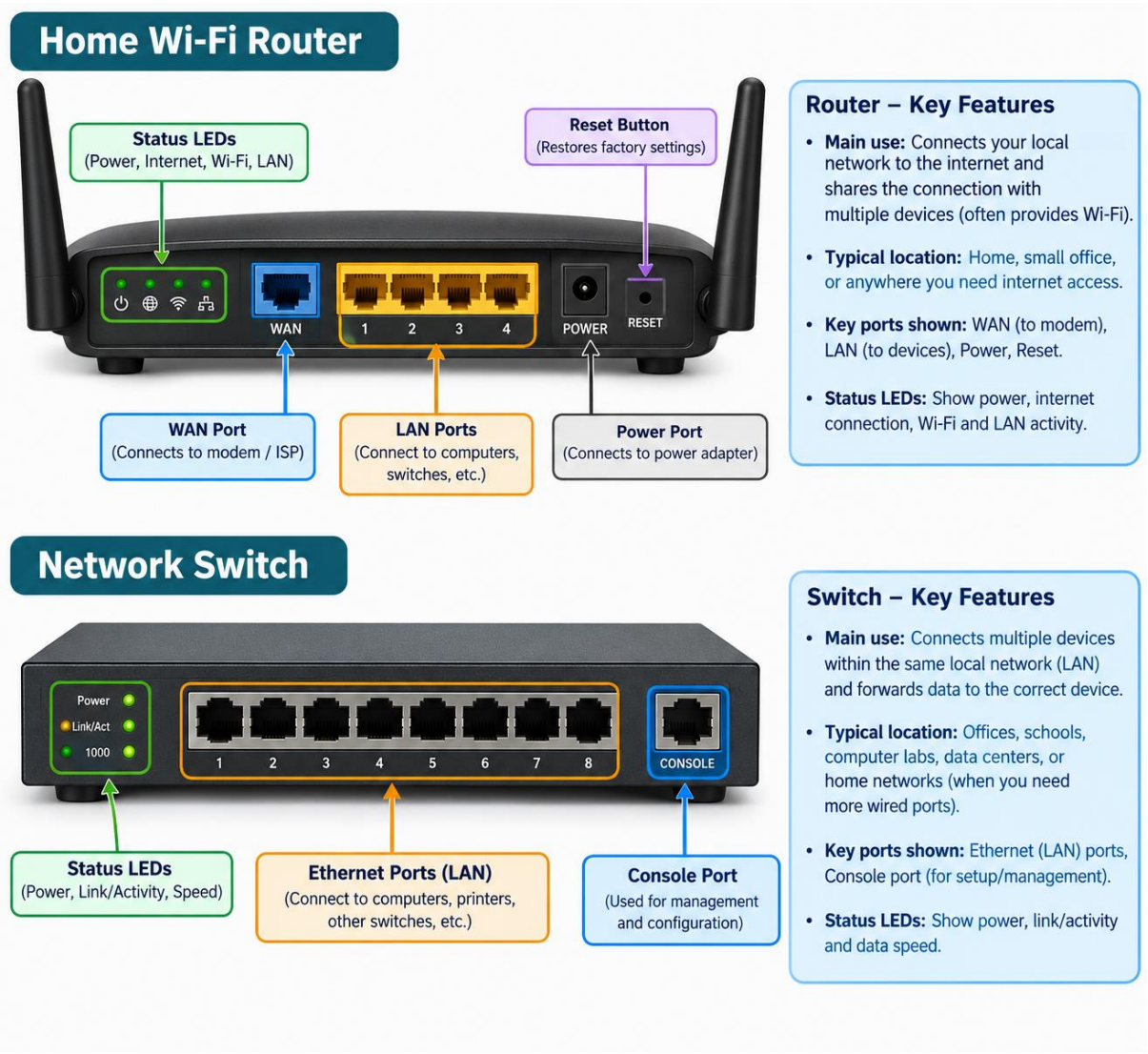
Assignment-4.2

TASK-1

Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



TASK-2

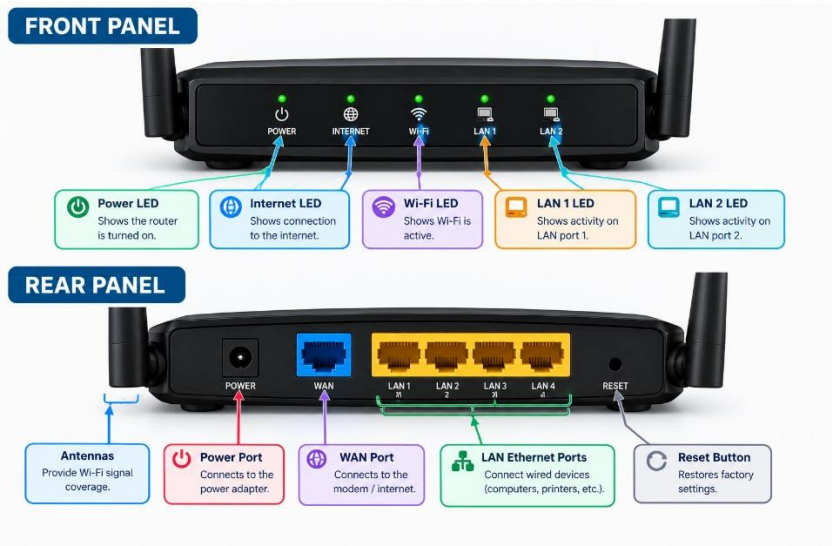
List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Feature	Routers	Switches
Network Type	Connects different networks together (such as linking your home local network to the global Internet).	Connects devices within a single network (such as linking computers, printers, and servers in the same building).
Addressing Used	Uses IP addresses to route data packets between different networks.	Uses MAC addresses to switch data frames between specific local devices.
Traffic Handling	Inspects the destination IP address and uses routing tables to find the best path to send data across separate networks.	Inspects the destination MAC address and looks up its MAC address table to forward data directly to the correct physical port.

TASK-3

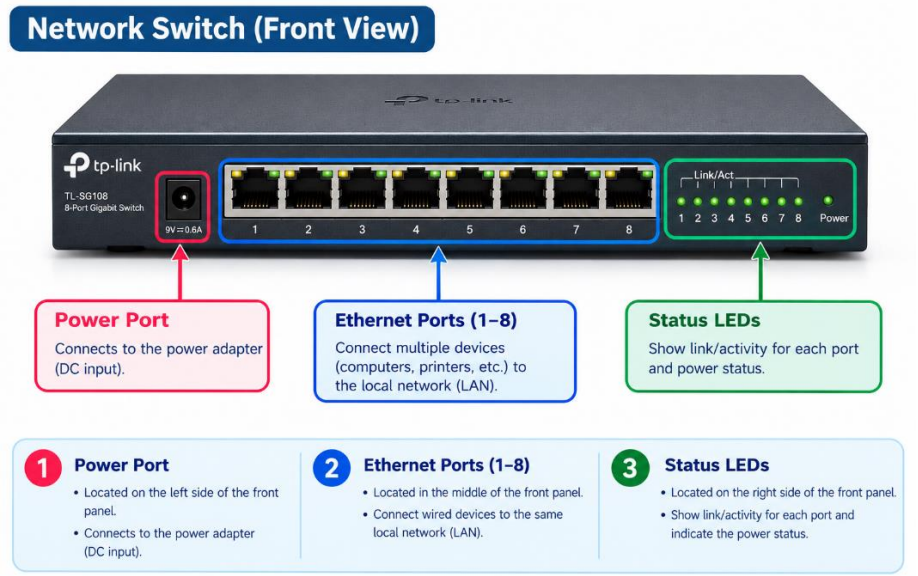
Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

1. Home Wi-Fi Router



Part	Position	Purpose
Power port	Back panel, left side	Connects the power adapter to the router.
WAN/Internet port	Back panel, near the power port	Connects the router to the modem or internet source.
LAN Ethernet ports (1–4)	Back panel, beside the WAN port	Connect wired devices such as computers and printers.
Status LEDs	Top/front panel	Indicate power, internet, Wi-Fi, and LAN activity.

2. Network Switch



Part	Position	Purpose
Power port	Left side of the pictured panel	Connects the switch to its power supply.
Ethernet ports (1–8)	Front panel, middle	Connect multiple wired devices to the same local network.
Link LEDs	Near the Ethernet ports	Show whether a connection is active and whether data is being transferred.

TASK-4

Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Scenario: Setting Up a Small Gaming Café Network

I am setting up a small gaming café for IPL streaming and multiplayer gaming with eight gaming PCs.

First, I would connect the router to the ISP line through the modem or Modem (depending on the internet connection). The router manages network traffic, assigns IP addresses, and provides firewall protection.

Next, I would connect the router's LAN port to an 8-port or larger Gigabit Ethernet switch using an Ethernet cable. Then, I would connect all eight gaming PCs to the switch using separate Ethernet cables.

Justification

- **Router first:** Connects the local network to the internet and manages network traffic.
- **Switch for eight PCs:** Provides enough wired ports to connect all gaming computers.
- **Ethernet cables:** Offer reliable, low-latency connections for multiplayer games and stable IPL streaming.
- **Gigabit switch:** Supports fast data transfer between connected devices**.**